Audio Equipment and Systems

Bachelor's Degree in Audiovisual Systems Engineering

Course overview and Introduction
2026-27

Engineering School
Department of Information
and Communication Technologies

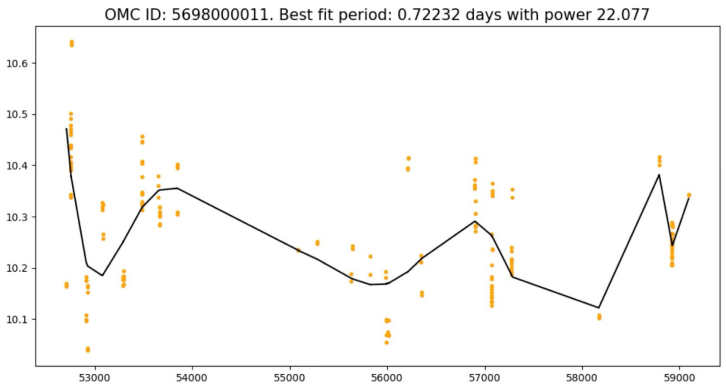
Universitat Pompeu Fabra Barcelona

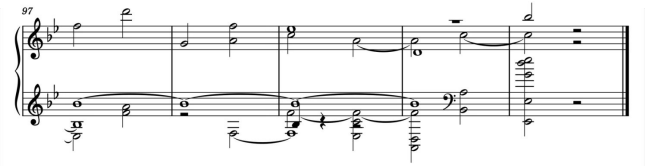
Adrià García Riber adrian.garcia@upf.edu
Xavier Lizarraga Seijas xavier.lizarraga@upf.edu

AES - TEAM

ADRIAN GARCIA RIBER

- **About me**
  - Associate Professor at UPF
  - Ph.D. in Music and Technology
  - Head of Audiovisuals at ESMUC
  - Sonification Researcher at Spanish Virtual Observatory (CAB-INTA-CSIC)





- **Areas of interest**
  - Sonification of astronomical data
  - AI and Machine learning
  - Astronomy and Astrophysics
  - Recording techniques
  - Audiovisual Postproduction
  - A/V Equipment and Installations

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esmuc | Escola Superior de Música de Catalunya

CENTRO DE ASTROBIOLOGÍA
ASOCIADO AL NASA ASTROBIOLOGY INSTITUTE
CSIC INTA

SVO Spanish Virtual Observatory

AES - TEAM

XAVIER LIZARRAGA

- **Dedication**
  - Co-founder and Lead Developer at Audio Merge
  - Researcher at Music Technology Group
  - Ex-coordinator at Audio Equipments and Systems (UPF school)
  - Member of Barcelona Music Tech Hub (BMTH)
- **Experience**
  - Sound Engineering
  - Music Live Performance & studio
  - Music Apps - Patchworks (SP)
  - Speech Technology - Verbio (SP)
  - Audio 3D - Auro Technologies (BE)
  - Audio Gear - Audio Merge (SP)
- **Interests**
  - Analog Gear Preservation
  - Audio Gear Analytics
  - Music Production Tools
  - Sound & Music Classification
  - Automatic Mixing & Mastering
  - Entrepreneurship

AUDIO MERGE

MTG Music Technology Group

ESSENTIA

xlizarraga@audio-merge.com
xavier.lizarraga@upf.edu
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D-Roy Station
Dub Producer

Calendar - Theory

Date	Reference and Professor	Room*	Topic
30/09/2026	T0 (Xavier & Adrià)	52.221	AES Course overview and Introduction
01/10/2026	T2 (Xavier)	54.004	Electronics - Review
05/10/2026	T1 (Adrià)	52.321	Sound and acoustics - Review
14/10/2026	T3-1 (Adrià)	52.329	Sound Capture and Microphones I
19/10/2026	T3-2 (Adrià)	52.S29	Sound Capture and Microphones II
26/10/2026	T4-1 (Adrià)	52.S29	Sound playback and Loudspeakers I
02/11/2026	T4-2 (Adrià)	52.321	Sound playback and Loudspeakers II
09/11/2026	T5 (Xavier)	52.S29	Transmission and Impedance Matching
16/11/2026	T6 (Xavier)	52.S29	Preamps and Power amplifiers
23/11/2026	T7 (Xavier)	52.S29	Analog Processing
30/11/2026	T8 (Xavier)	52.321	Analog Playback and Recording
02/12/2026	Q&A (Xavier & Adrià)	52.321	Course Review

* Final rooms may change. Consult the calendar.

Recommended Bibliography

- Everest, F. A. (2022). Master handbook of acoustics.
- Eargle, J. (2012). The Microphone Book: From mono to stereo to surround-a guide to microphone design and application. Routledge.
- Ballou, G. (2013). Handbook for sound engineers. Taylor & Francis.
- Eargle, J. (2003). Loudspeaker handbook. Springer Science & Business Media.
- Kadis, J. (2012). The science of sound recording. Routledge.
- Brixen, E. (2020). Audio metering: measurements, standards and practice. Focal Press.

UPFinder link

Calendar - Labs and Seminars

Date	Reference and Professor	Room*	Topic
07/10/2026	Seminar 1** (Xavier & Adrià)	52.221	Basic exercises related to T1 and T2
08/10/2026	P1 (P101-Xavier/P102-Adrià)	54.025	Introduction to AES Measures
15/10/2026	P2	54.025	Microphones I
21/10/2026	Seminar 2 (S101-Xavier/S102-Adrià)	52.329 52.105	Microphones exercises
22/10/2026	P3	54.025	Microphones II
29/10/2026	Seminar 3	52.105	Loudspeakers exercises
05/11/2026	P4	54.025	Loudspeakers I
12/11/2026	P5	54.025	Loudspeakers II
19/11/2026	Seminar 4	52.105	Power amplifiers exercises
26/11/2026	P6	54.025	Dante & Pro Tools
03/12/2026	Seminar review**	52.S31	Final exam exercises

* Final rooms may change. Consult the calendar.
** Only the first and the last seminars will take place with all the students together (Theory format). The rest of the seminars will take place with students divided by groups (Lab format).

Spaces - Rooms - Equipment

- Theory | Seminars
  - Roc Boronat Building (52)

https://www.upf.edu/en/web/campus/roc-boronat

- Labs
  - Tallers Building (54)

https://www.upf.edu/web/campus/area-tallers

    - Audio Lab (54025)
    - Close to the Local Networks Lab
- Equipment:
  - Cables, connectors, mixing console, microphones, loudspeakers
  - Acoustic treatment solutions
  - Audio Interfaces:
    - Focusrite Scarlett 2i2 [features]
- Softwares:
  - REW, REAPER and AUDACITY
- Coding:
  - Python - Jupyter notebooks

P1- Introduction to AES Measures

In this session, we will learn about:

- REW settings, menus and displays for transfer function analysis
- Different test signals and methods
- Introducing the logarithmic sweep tone technique
- How to measure audio equipments with REW using a bi-channel comparison



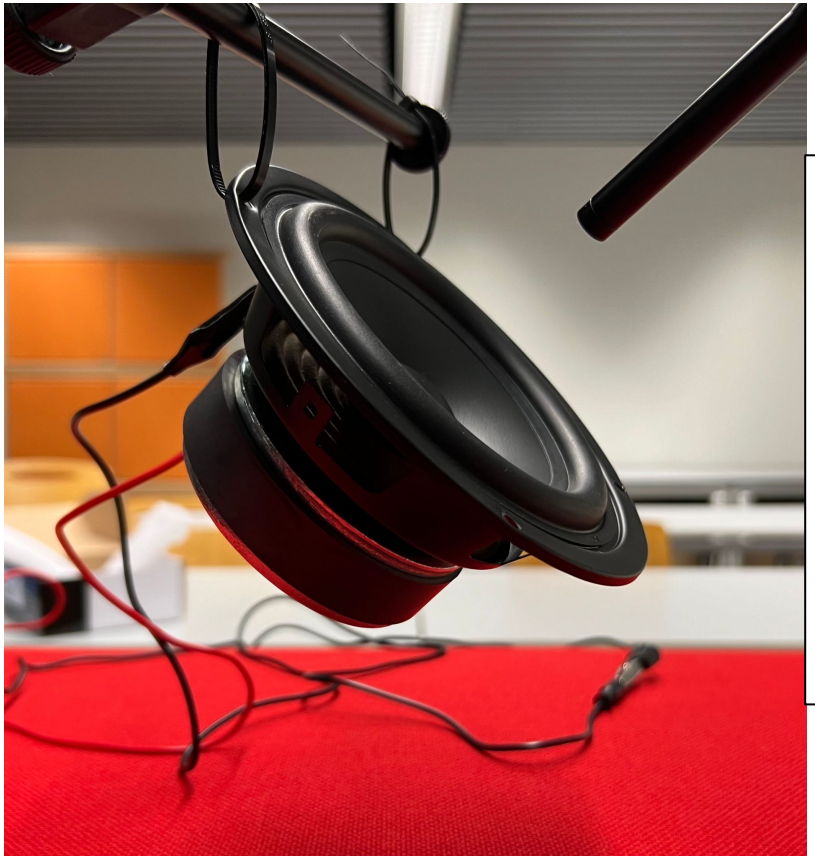
P2&P3 Microphones

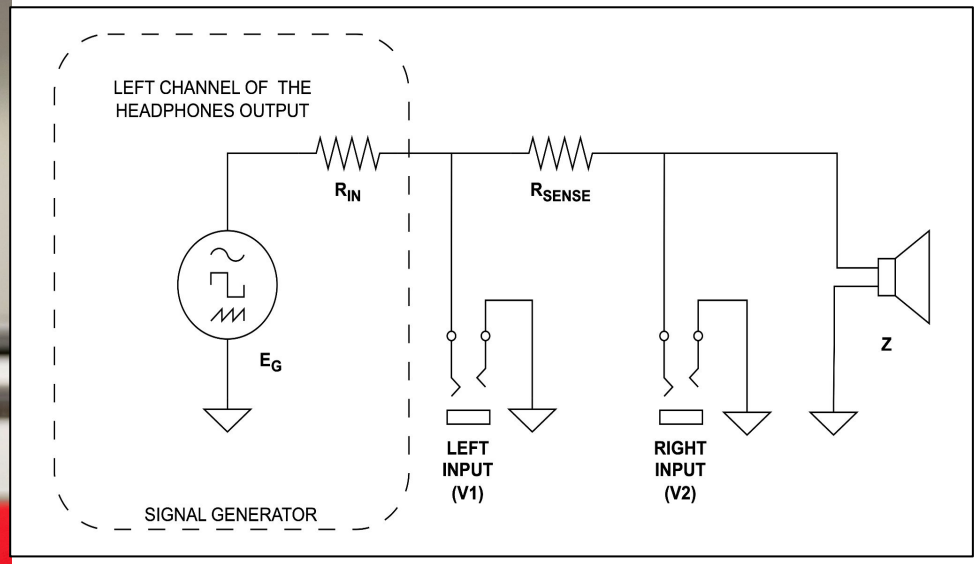
These labs aim to experimentally determine the technical specifications of dynamic and electrostatic microphones, such as the frequency response, proximity effect and directivity.



P4&P5 Loudspeakers & Impedances

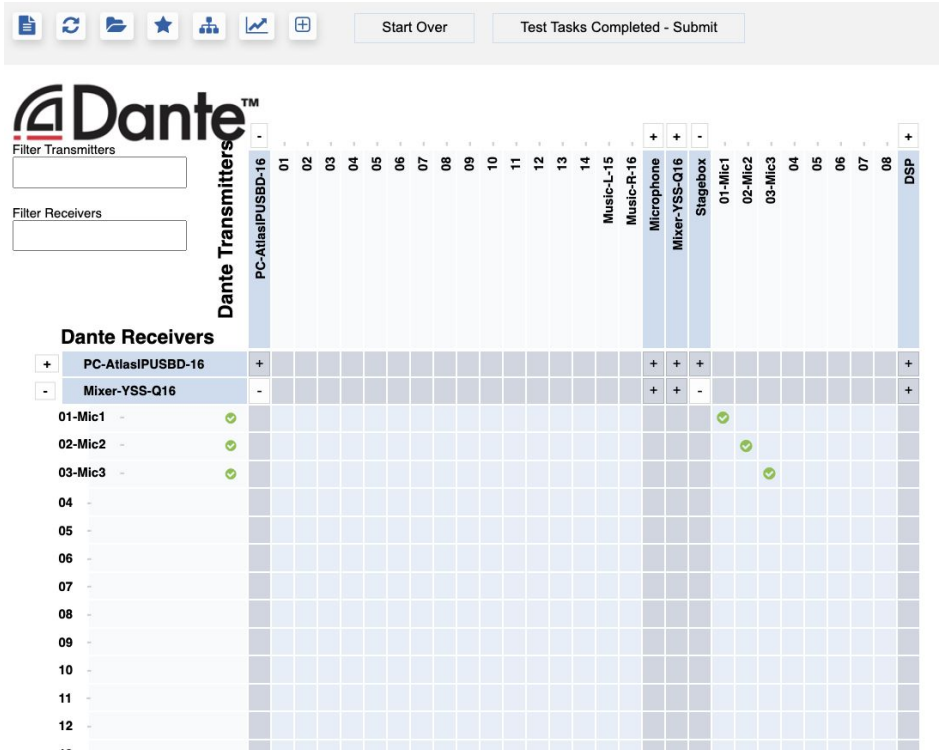
The goal of these labs is to experimentally determine the technical specifications of loudspeakers, analyzing the influence of the enclosing box, the crossover zone, directivity and electric impedance.





P6 - Dante|ProTools|Audio plugins

In this Lab we will work with *Dante* and *Pro Tools*, with a special focus on routing and processing, to offer an overview of the workflow and possibilities of these professional environments commonly used on sound reinforcement, recording, mixing, and mastering processes. If we have time, we will also introduce the use of *JUCE* in the development of audio plugins.



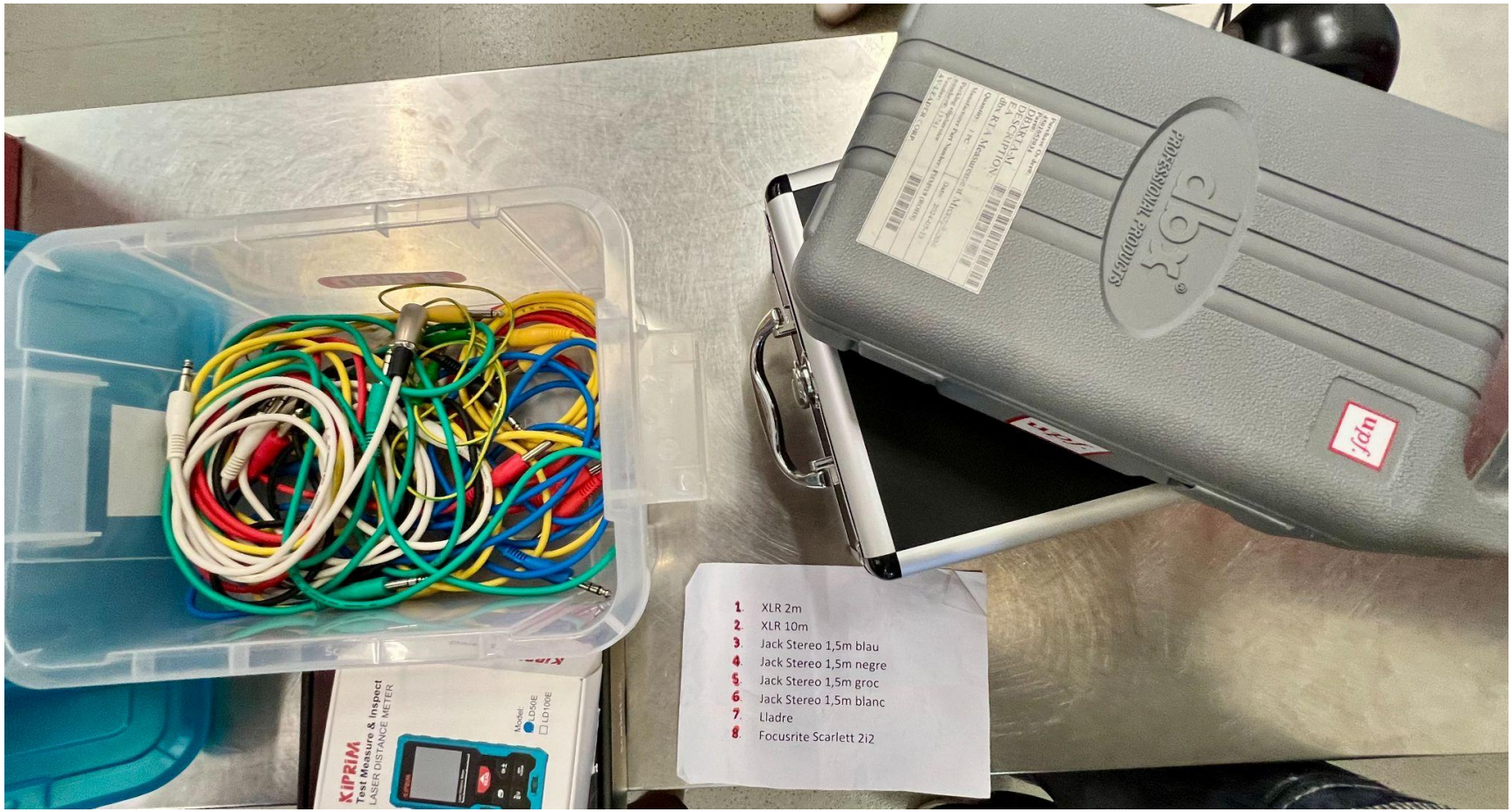




Deliverables - Labs and Seminars

Delivery Date	Reference	Topic
21/10/2026	S1	Acoustics and electronics - Exercises
22/10/2026	P1	Introduction to AES Measures
29/10/2026	P2	Microphones I
04/11/2026	S2	Microphones - Exercises
05/11/2026	P3	Microphones II
12/11/2026	S3	Loudspeakers - Exercises
19/11/2026	P4	Loudspeakers I
26/11/2026	P5	Loudspeakers II
03/12/2026	S4	Power Amplifiers - Exercises
10/12/2026	P6	Dante & Pro Tools

AES Lending Service - At the Library



Lab Groups - Assignation

- 5-6 students for each LAB group
- LAB1 will be next week
  - So, please create a group before next Tuesday
  - Otherwise we will make it randomly
- **Please, fill your group in the XLS file** https://docs.google.com/spreadsheets/d/1kLbQJVMCZfa4ZdooRlLUixXNzPbvuEgBk8li4OhIkM8/edit?usp=sharing
- NO LAB group changes are allowed
- If you have any questions, contact us!

Evaluation

The final evaluation of the course consists of **three parts**:

1. A final examination at the end of the trimester (including Lectures and Seminars)
2. Six laboratory reports derived from the practical sessions.
3. Some seminar exercises derived from the theory sessions.

To be evaluated at the end of the trimester, it is necessary to deliver all the laboratory reports and to do the final exam.

1. The theory test, consisting of questionnaire questions and exercises, will account for 50% of the final grade.
2. The practical exercises carried out in groups during the course will account for 40% of the final grade.
3. The seminar exercises will account for the remaining 10% of the final grade.

To obtain a final grade, it is mandatory to obtain a grade higher than 4 in the final exam and to deliver all the reports.

On July there will be another opportunity for those not passing the evaluation.

To attend this extraordinary call it is mandatory to have previously submitted all the laboratory reports.

Contact details
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Adrià García Riber adrian.garcia@upf.edu room: 55.307

Final Exam

December 18th at 15:00.

Please let us know if you have any **special educational needs (NEE)**.

- The test is graded on a scale of 100 points.
- It is divided into two parts:
  - Physics - Acoustics - Microphones - Loudspeakers
  - Electronics - Audio Transmission - Impedance - Amplifiers
- Each part is split by:
  - Theory
  - Exercises
- It usually includes LABs and SEMINAR exercises.
- Calculator will be allowed but formulas and equations are not provided.
- There is no intermediate exam and a lot of content, so it is recommended to attend the theory classes.

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Final Degree Projects

In previous years, we have supervised some final degree projects about:

- Audio Equipment Measurements
- Virtual Analog Modelling
- Auralization
- Real-Time Audio Processing
- Music Genre Classification
- Sound Event Detection
- Low-level Sound Descriptors
- Audio Gear / Electronics

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Plagiarism – warning and advice

- Plagiarism is the practice of appropriating the words or production of another as one's own. It is considered malpractice and it is both illegal and unethical.
- Examples include: identity theft; use of forbidden material, including apps and devices, during tests/exams/academic works; assisting someone in copying tests/exams/academic works.
- The School has a commission aimed at dealing with issues related to plagiarism.
- Types of offences: minor, serious, very serious (applied to TFGs and TFMs)
  - Specific sanctions apply in each case: warning, file open (recording the offence), fail, including loss of right to the retake in July, etc.
- Guidelines:
  - https://guiesbibtic.upf.edu/treball-academic
  - https://guiesbibtic.upf.edu/dretsautor/plagi
- How to avoid it:
  - https://www.youtube.com/watch?v=PzZsButRaHs

Plagi – advertiment i consells

- El plagi és la pràctica d'apropiar-se de les paraules o la producció d'un altre com a pròpies. Es considera una mala praxis i és tant il·legal com poc ètic.
- Exemples inclouen: robatori d'identitat; ús de material prohibit, incloses aplicacions i dispositius, durant proves/exàmens/treballs acadèmics; ajudar algú a copiar proves/exàmens/treballs acadèmics.
- La nostra escola compta amb una comissió d'integritat acadèmica destinada a tractar temes relacionats amb el plagi.
- Tipus d'infraccions: lleus, greus, molt greus (aplicades a TFG i TFM)
  - S'apliquen sancions específiques a cada cas: amonestació, obertura d'expedient (inscripció de la infracció), suspens, inclosa la pèrdua del dret a la repetició al juliol, etc.
- Guies disponibles:
  - https://guiesbibtic.upf.edu/treball-academic
  - https://guiesbibtic.upf.edu/dretsautor/plagi
- Com evitar-ho
  - https://www.youtube.com/watch?v=9rwtOwGwVRU&t=566s

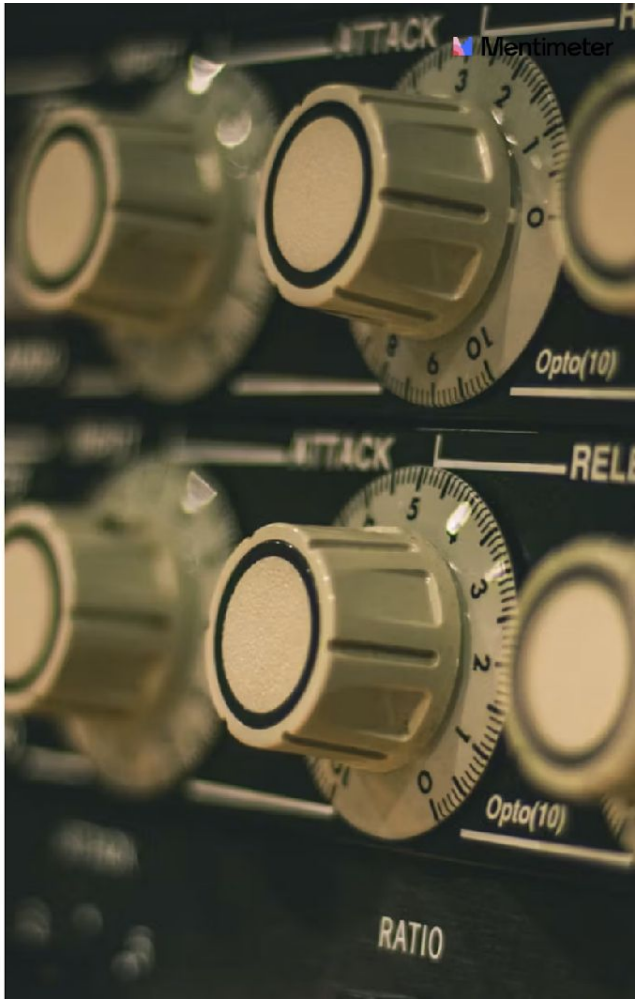
Plagio – advertencia y consejos

- El plagio es la práctica de apropiarse de las palabras o la producción de otro como propias. Se considera una mala praxis y es tanto ilegal como poco ético.
- Ejemplos incluyen: robo de identidad; uso de material prohibido, incluidas aplicaciones y dispositivos, durante pruebas/exámenes/trabajos académicos; ayudar a alguien a copiar pruebas/exámenes/trabajos académicos.
- La comisión de integridad académica tratar los temas relacionados con el plagio
- Tipos de infracciones: leves, graves, muy graves (aplicadas a TFG y TFM)
  - Son específicas en cada caso: amonestación, apertura de expediente (inscripción de la infracción), suspenso, incluida la pérdida del derecho a la repetición en julio, etc.
- Guías disponibles:
  - https://guiesbibtic.upf.edu/treball-academic
  - https://guiesbibtic.upf.edu/dretsautor/plagi
- Como evitarlo
  - https://www.youtube.com/watch?v=9rwtOwGwVRU&t=566s

Welcome questionnaire

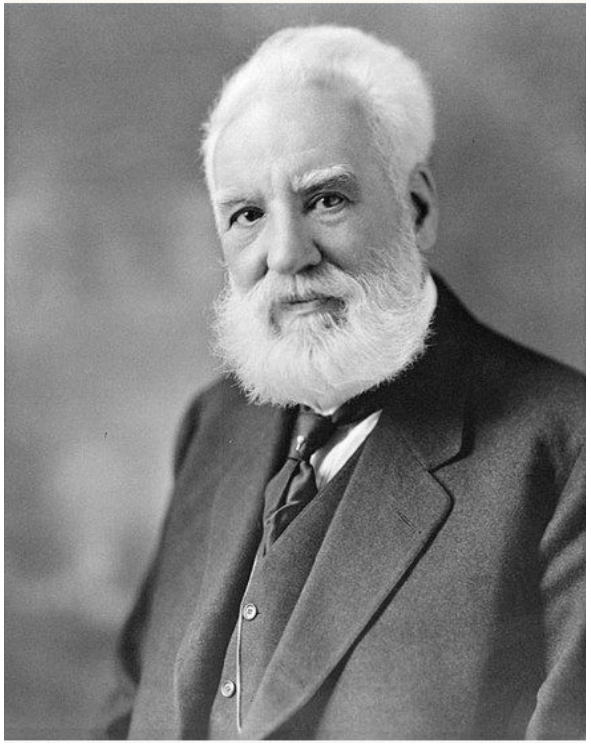
Audio Equipment and Systems

Welcome and expectations



Introduction | Historical Review

- Traditionally, the first audio companies have emerged from **scientific research** and **patents**.
- US, UK and Germany **researchers** were very active at that time.
  - **Bell**, **Hertz, Hass** and **Blumlein** are names of the researchers or inventors.



Alexander Graham Bell [source]

Introduction | Historical Review

- Audio companies started focused on communication and media broadcasting systems.
- **Telegraph, telephone**, **filming**, **radio**, and **music** later.



W7DET Seattle Radio Station, 1957 [source]

Introduction | Historical Review

US companies

- Bell Telephone Company
- American Telephone and Telegraph (AT&T)
- New England Telephone and Telegraph
- General Electric
- Western Electric
- Westinghouse
- RCA
- Altec
- Universal Audio (UREI)
- Pultec

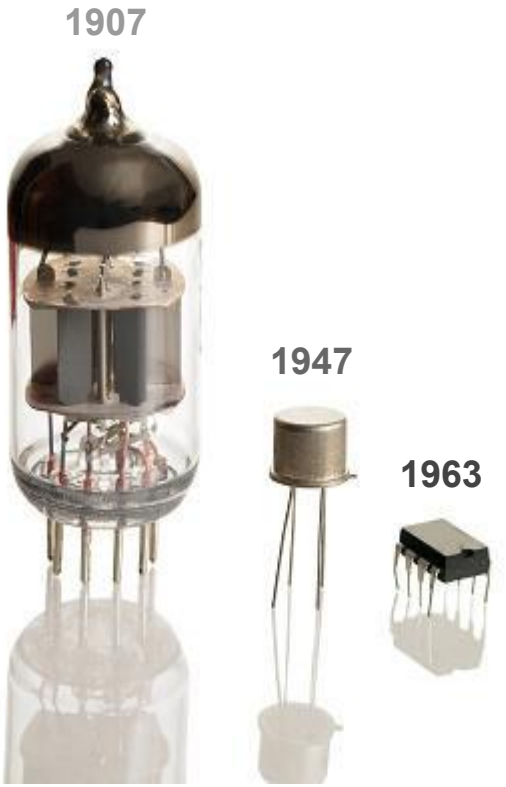
European companies

- Thetrophone (FR)
- Electrophone (UK)
- Telefunken (DE)
- Neumann (DE)
- Spanish Wireless Telegraph and Telephone Corporation (SP)
- EMI (UK)
- British Broadcasting Company - BBC (UK)

https://www.soundandcommunications.com/industry-pioneers/

Introduction | Historical Review

- AT has been connected to the development of **electronics components**.
- *Vacuum tubes*, *transistors* and *OPAMPs* have influenced deeply the audio tech field.



Valve, transistor and opamp [source]

Introduction | Motivation

- Professional musicians, producers and engineers are in love with the tools exploited by the pioneers of audio equipment and systems.



Ray Charles



Geoff Emerick - Beatles mixing engineer

Introduction | Motivation

- Professional musicians, producers and engineers are in love with the tools exploited by the pioneers of audio equipment and systems.



Les Paul and Mary Ford



Robert Moog at Moog Synthesizers

Introduction | Motivation

- History of compressors [link]
- History of the studio console (part1) [link]
- History of the studio console (part2) [link]
- History of the studio EQ [link]

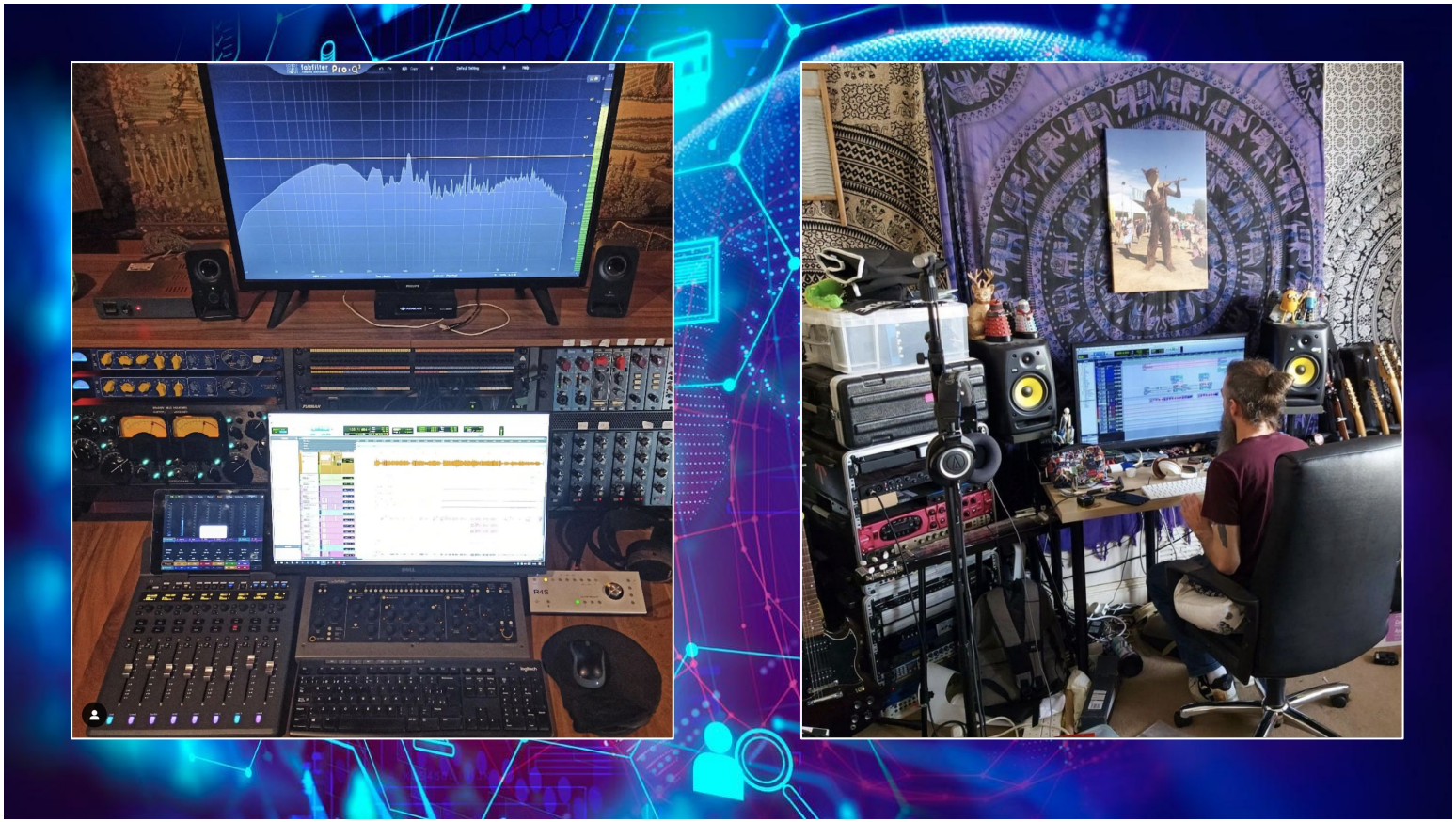


Introduction | Motivation



Introduction | Motivation

- Nowadays, it seems digital applications should replace equipments but they co-live in hybrid systems. Audio analog equipment still realm the field.



Introduction | Motivation

- Nowadays, it seems digital applications should replace equipments but they co-live in hybrid systems. Audio analog equipment still realm the field.



Grammy winner producer, Jack Joseph Puig

Introduction | Motivation

- Much of the market in digital music production tools is focused on emulations of analog equipment recognized for its *sound signature*.



Universal Audio 1176LN - Classic FET Compressor [url]



UA 1176 Classic Limiter Collection [url]



Waves CLA-76 Compressor-Limiter [url]

AES Scenarios



AES Scenarios
Live Music Performance





AES Scenarios
Music Studio



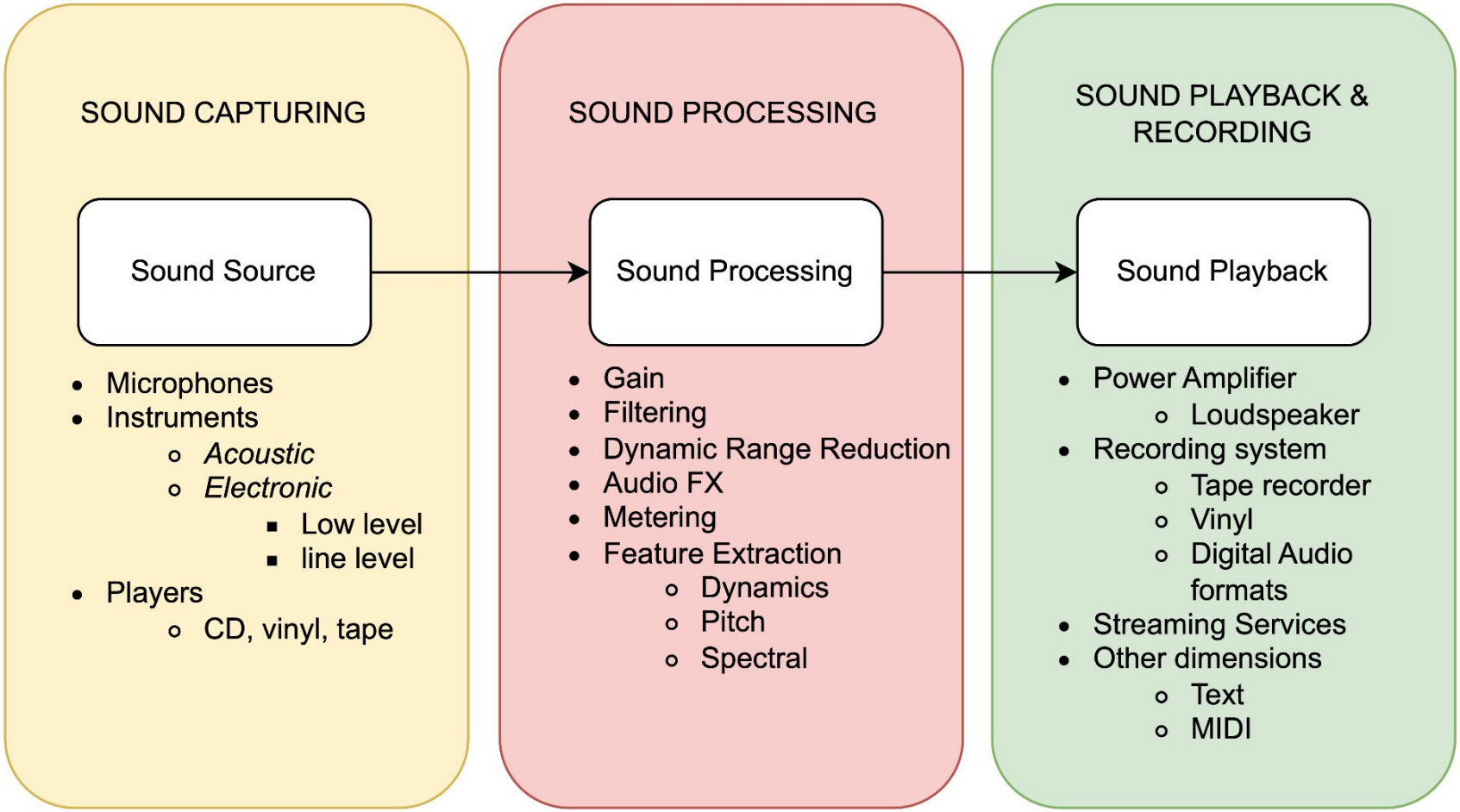
AES Scenarios
Radio Station



AES Scenarios
Smartphone

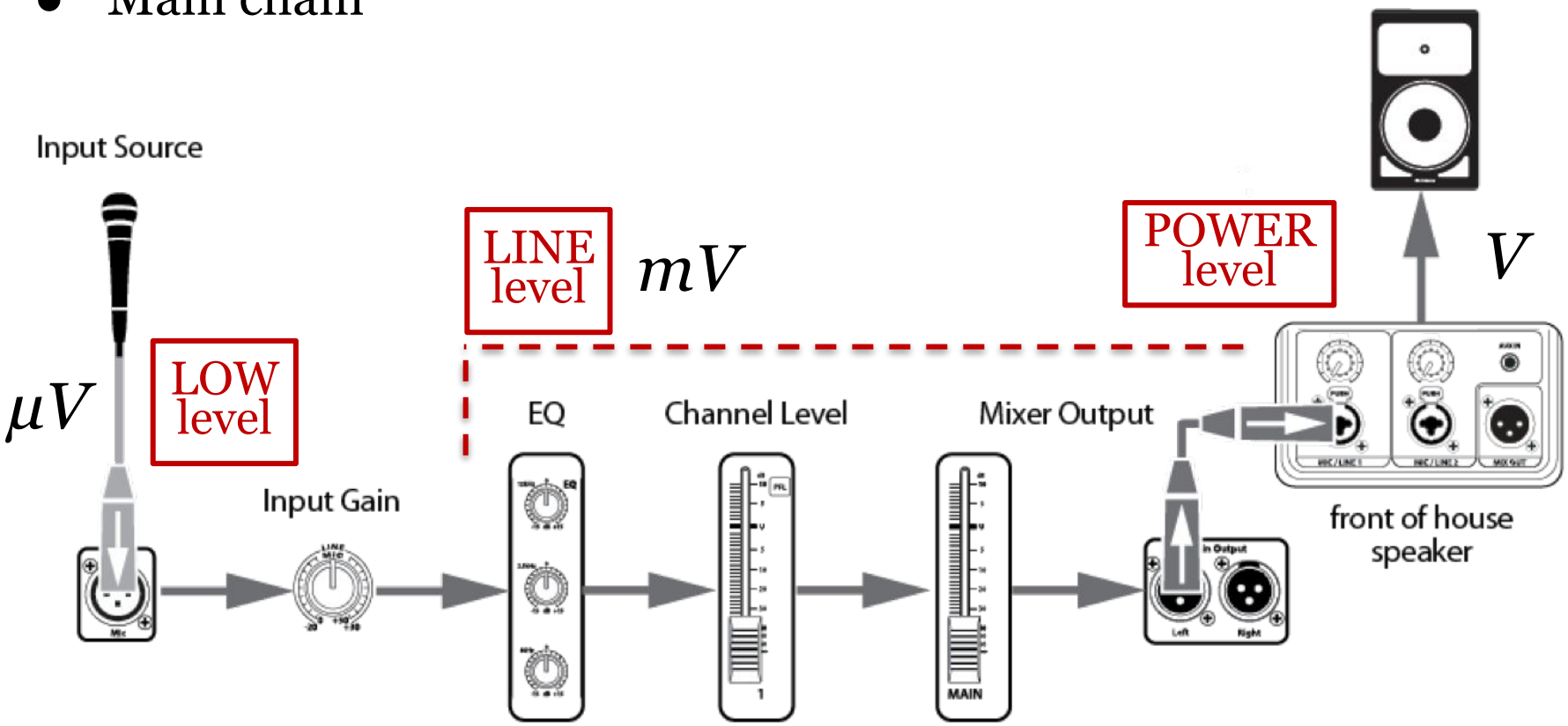


AES Diagram


SOUND CAPTURING
Sound Source
Microphones
Instruments
Acoustic
Electronic
Low level
line level
Players
CD, vinyl, tape
SOUND PROCESSING
Sound Processing
Gain
Filtering
Dynamic Range Reduction
Audio FX
Metering
Feature Extraction
Dynamics
Pitch
Spectral
SOUND PLAYBACK & RECORDING
Sound Playback
Power Amplifier
Loudspeaker
Recording system
Tape recorder
Vinyl
Digital Audio formats
Streaming Services
Other dimensions
Text
MIDI

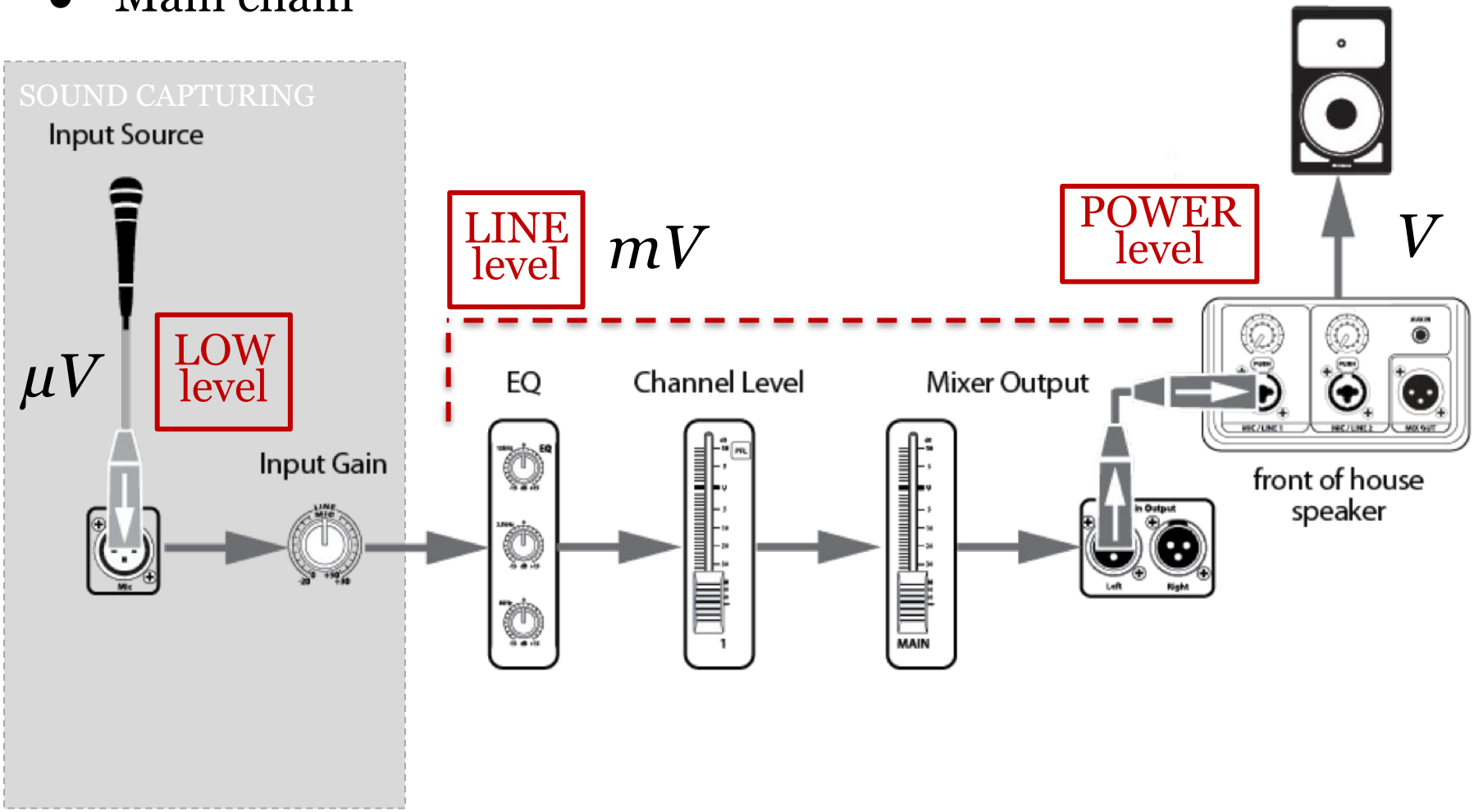
Audio Equipments and Systems

- Main chain



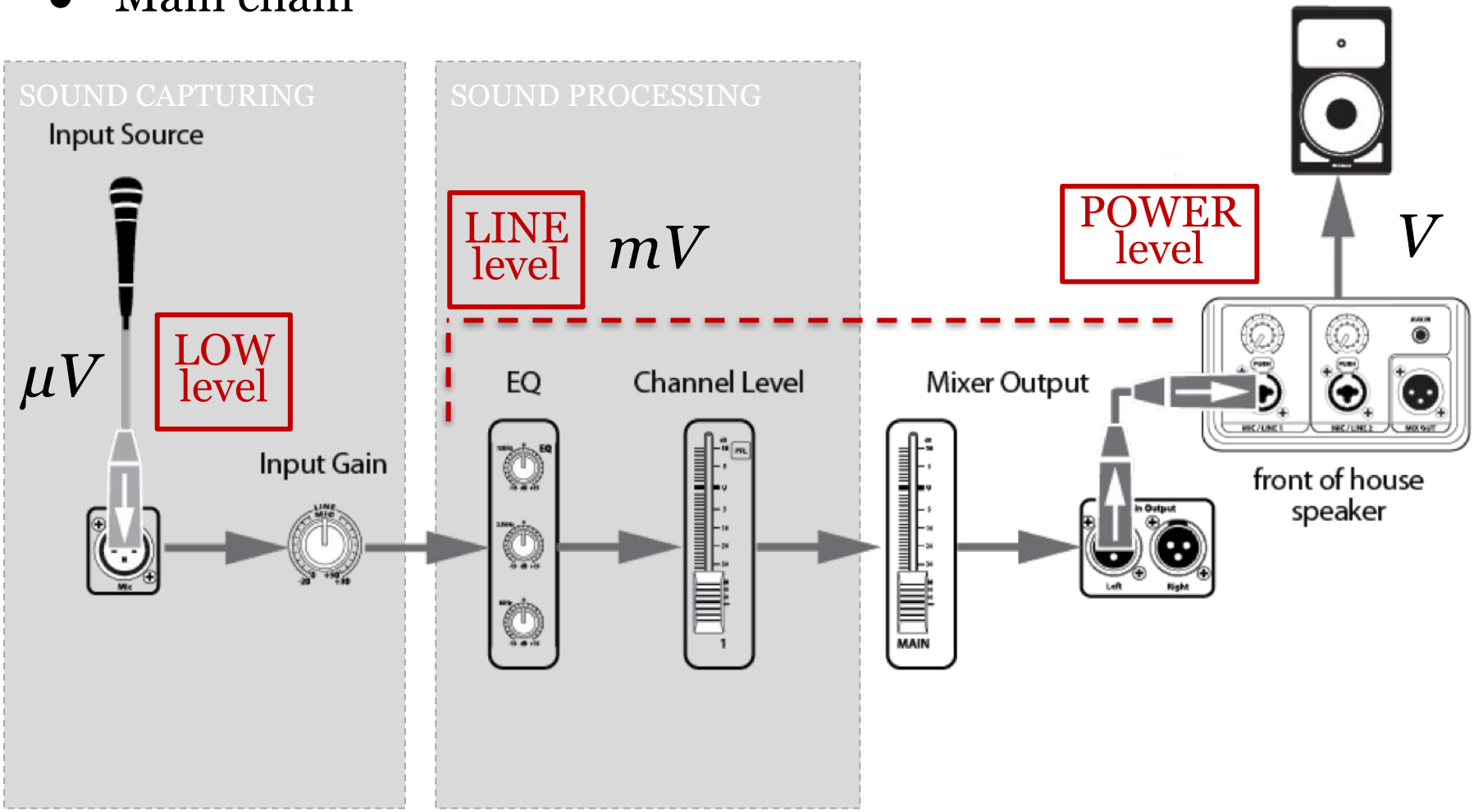
Audio Equipments and Systems

- Main chain



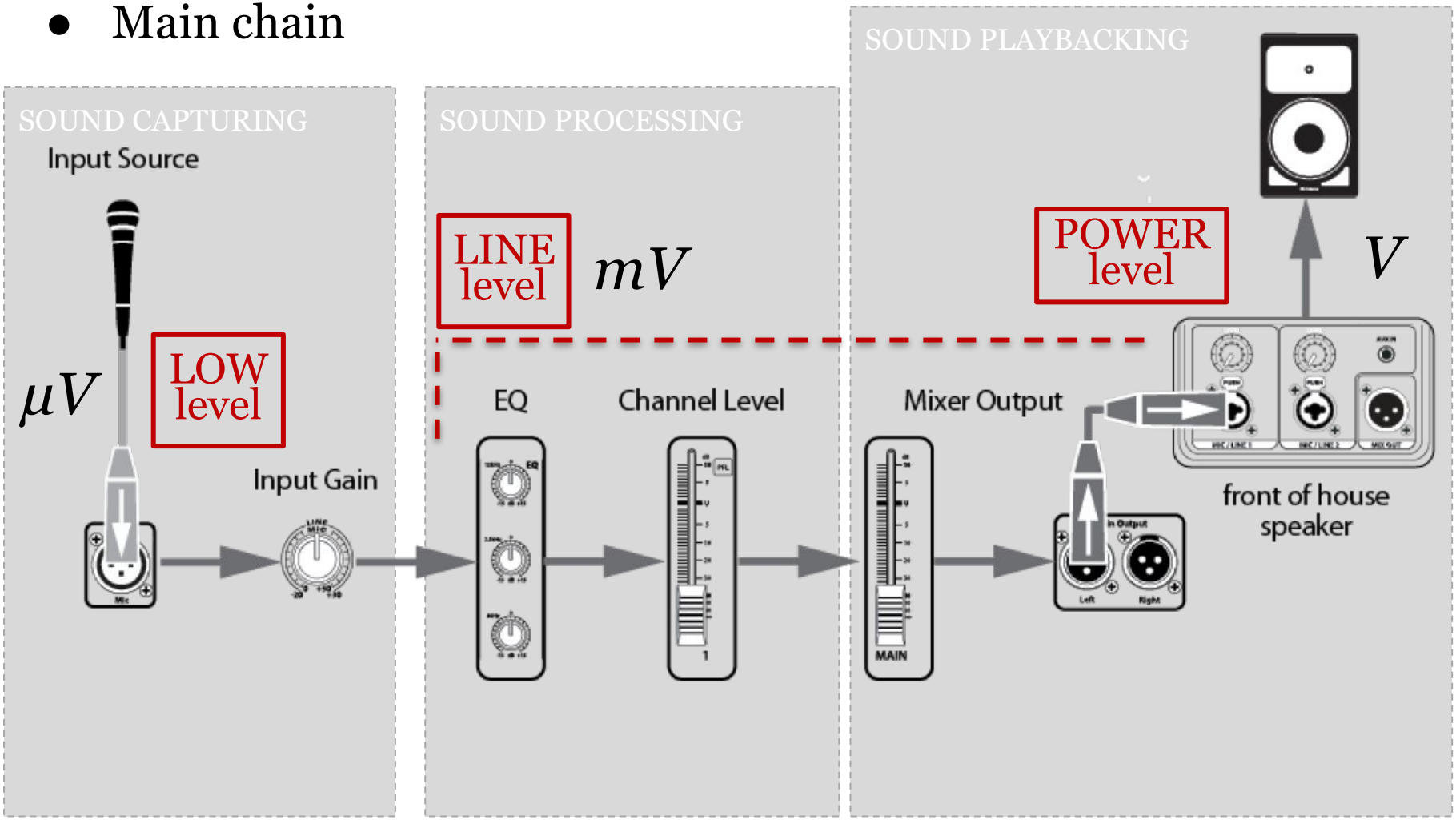
Audio Equipments and Systems

- Main chain



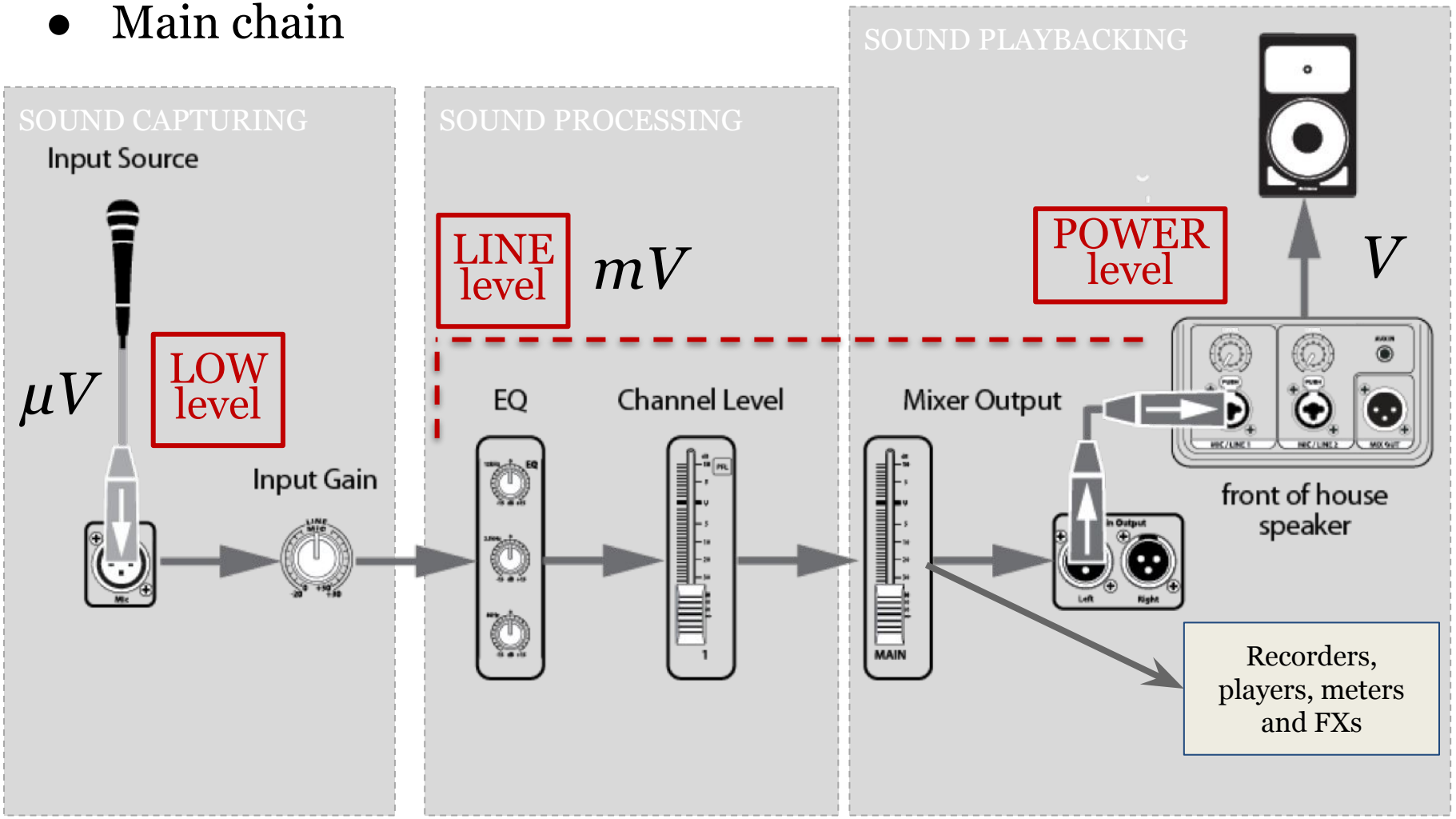
Audio Equipments and Systems

- Main chain



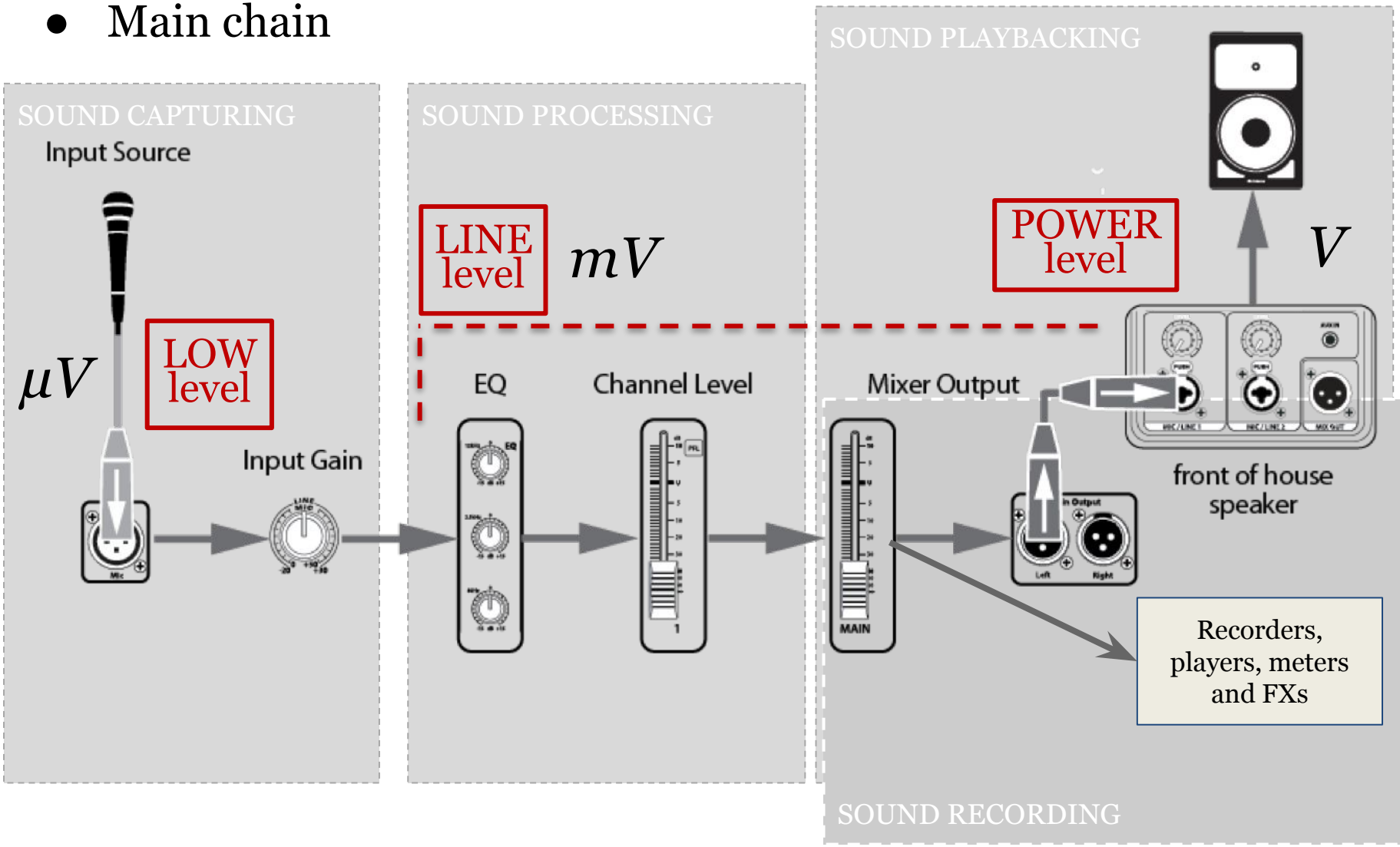
Audio Equipments and Systems

- Main chain



Audio Equipments and Systems

- Main chain




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